# Decision-Making Analysis

The Case of Delta Tech Pvt. Ltd., Biratnagar

## Group Assignment & Task Allocation

| Member | Focus Area | Target Word Count |
|---|---|---|
| Member 1 | Foundation & Decision Analysis (Case Overview, Programmed & Non-Programmed Decisions) | ~550 words |
| Member 2 | Environment Conditions & Intuitive Judgment (Decision-Making Condition, Role of Intuition/Heuristics) | ~550 words |
| Member 3 | Rational Process Application & Synthesis (Rational Decision-Making Model, Conclusion & Recommendations) | ~650 words |

# MEMBER 1 — Foundation & Decision Analysis

*Focus: Context setting and core identification of decision types (~550 words)*

## 1. Introduction & Case Overview

Delta Tech Pvt. Ltd. is a small technology firm based in Biratnagar, Nepal, that has set itself the objective of designing and launching a mobile application aimed at helping local retailers digitize and streamline their business operations. As a growing city with an expanding small-business sector, Biratnagar presents a promising but largely untested market for such a product. However, Delta Tech's leadership faces four interlocking constraints that shape every decision on this project: a limited budget that restricts how much can be spent on development, testing, and marketing; uncertain customer demand, since no reliable data exists on whether local retailers will adopt a digital tool of this kind; a shortage of locally available technical skills, particularly experienced mobile app developers; and a tight project timeline that leaves little room for trial and error. Together, these constraints mean that Delta Tech's managers must make a series of decisions that range from routine and repetitive to highly uncertain and strategic in nature. Distinguishing between these two broad categories of decisions—programmed and non-programmed—provides a useful starting point for analyzing how the company should organize its decision-making process going forward.

## 2. Programmed Decisions

Programmed decisions are structured, repetitive, and routine in nature. They arise from situations that occur frequently enough that an organization can develop standard operating procedures, rules, or policies to handle them efficiently, without needing fresh analysis each time. Because the problem is well understood and has occurred before in a similar form, managers can rely on established guidelines rather than deliberating from scratch. Within the Delta Tech case, several recurring operational matters fall squarely into this category:

- **Standard payroll and contract templates:** Whenever Delta Tech brings on new hires, payroll processing and employment contract templates can follow a fixed, pre-established format, requiring little managerial judgment.
- **Routine stand-ups and code check-ins:** Daily or weekly project stand-up meetings and version-control check-in procedures are repetitive workflow activities that the development team can run through a consistent, agreed-upon routine.
- **Software licensing and hosting renewals:** Recurring subscriptions for development tools, software licenses, or cloud hosting can be renewed automatically or through a simple standing approval process, since the decision criteria (cost, renewal date, continued need) rarely change.

By handling these matters through routine procedures, Delta Tech's managers free up their limited time and cognitive energy to focus on the more complex, judgment-intensive problems the project presents.

## 3. Non-Programmed Decisions

Non-programmed decisions are unique, unstructured, and poorly defined. They typically involve novel or unusual situations for which no ready-made procedure exists, meaning managers must rely on judgment, creativity, and careful analysis rather than routine. Delta Tech's core project challenges are almost entirely of this type, since the company has no prior experience launching a similar app in this specific market:

- **Budget allocation between development and marketing:** With limited funds, deciding how much to invest in building the app versus promoting it to local retailers is a genuinely open-ended trade-off with no formula to fall back on.
- **Pivot strategy if demand differs from expectations:** Because retailer demand in Biratnagar is largely unknown, management must be prepared to make an unstructured, judgment-based decision about how and when to pivot the product if early adoption patterns diverge from initial assumptions.
- **Sourcing or upskilling talent under time pressure:** Given the shortage of experienced developers locally, deciding whether to recruit externally, outsource, or rapidly train junior staff is a unique problem shaped by the specific constraints of cost, urgency, and available skill in Biratnagar.

These non-programmed decisions carry the greatest risk and require the most active involvement of senior management, since no past precedent can reliably guide the choice.

# MEMBER 2 — Environment Conditions & Intuitive Judgment

*Focus: Theoretical environmental framing and behavioral decision-making (~550 words)*

## 4. Decision-Making Condition Analysis (Question 2)

Decision-making conditions are generally classified into three categories: certainty, risk, and uncertainty. Certainty exists when a manager knows exactly what the outcome of each alternative will be, because complete and accurate information is available. Risk exists when the possible outcomes are known, but their probabilities must be estimated, usually from historical data or statistical patterns. Uncertainty exists when neither the outcomes nor their probabilities are clearly known, forcing managers to act on incomplete information, intuition, and best judgment.

Delta Tech's situation is best classified as a condition of **uncertainty, with elements of high risk**. This is not a condition of certainty, because management cannot predict with confidence how the app will perform once launched. It is also not a straightforward case of risk, because Delta Tech lacks the historical data that risk-based decision-making requires. There is no existing track record of mobile app adoption among local retailers in Biratnagar, so managers cannot assign reliable probabilities to different demand scenarios—they can only estimate broadly. Similarly, management does not know in advance whether it will be able to find or develop the right technical talent within the available time and budget, which introduces a further unknown into the equation. Because both the potential outcomes (market response, talent availability) and the likelihood of each outcome are largely unknown rather than statistically estimable, the overall environment is one of uncertainty. At the same time, the scale of the budget and time invested means that a wrong decision carries substantial financial and reputational consequences for Delta Tech, which is why the situation also carries a strong element of high risk layered on top of the underlying uncertainty.

## 5. The Role of Intuition and Heuristics (Question 4)

When managers face uncertainty and must act quickly, as Delta Tech's leadership must under its tight project timeline, they frequently turn to intuition and heuristics rather than exhaustive rational analysis. Intuitive judgment draws on a manager's accumulated experience, pattern recognition, and professional instinct to reach a workable decision quickly, even when complete data is unavailable. This is not a careless shortcut but a practical necessity when time and information are both scarce.

Two heuristics are particularly relevant to Delta Tech's decision process. The first is the **availability heuristic**, where managers lean on examples that come most readily to mind—such as recalling a past regional technology project that succeeded or failed—and use that memory as a mental shortcut for judging the likely success of the current app, even though the two situations may not be truly comparable. The second is **satisficing**, a concept introduced by Herbert Simon, in which decision-makers choose the first option that is "good enough" to meet minimum acceptable criteria, rather than continuing to search for the theoretically optimal choice. Given Delta Tech's limited budget and time, management is likely to satisfice by selecting a workable development framework or a reasonably qualified developer immediately available, instead of holding out for a perfect but slower or costlier alternative. Together, these heuristics allow Delta Tech's managers to move forward confidently despite incomplete information, though they also introduce the risk of bias if relied upon without any structured cross-check.

# MEMBER 3 — Rational Process Application & Synthesis

*Focus: Actionable framework application and report integration (~650 words)*

## 6. Step-by-Step Rational Decision-Making Process

Although Delta Tech operates largely under uncertainty, applying the classic rational decision-making model can still bring useful structure and discipline to its central dilemma: how to launch a viable mobile app on time and within budget despite talent shortages and unclear demand. The steps below apply this model directly to the case.

### Step 1 — Define the Problem

The core problem facing Delta Tech is how to design, build, and launch a functional mobile app for local retailers in Biratnagar within a constrained budget and a tight deadline, while dealing with uncertain market demand and a shortage of skilled developers.

### Step 2 — Identify Decision Criteria

The relevant criteria for evaluating any course of action include: overall cost, speed to market, likely user adoption among local retailers, and the technical scalability of the chosen solution.

### Step 3 — Allocate Weights to the Criteria

Given the severity of the time constraint, speed to market is weighted most heavily, followed closely by cost. Technical scalability, while important for the long term, is weighted lower for now, since an imperfect but functioning product that reaches the market on time is more valuable than a highly scalable one that arrives too late or over budget.

### Step 4 — Develop Alternatives

Management can consider several alternatives, including: (a) hiring expensive senior remote developers versus training local junior staff; and (b) building a full-featured application versus building a lean minimum viable product (MVP) focused on the retailers' most essential needs.

### Step 5 — Evaluate the Alternatives

Hiring senior remote developers would improve technical quality and speed but strains the limited budget, while training juniors is cheaper but slower and riskier given the tight timeline. Similarly, building a full-featured app better addresses long-term scalability but delays launch and raises cost, whereas an MVP can be delivered faster and more cheaply, allowing Delta Tech to test real retailer demand before committing further resources.

### Step 6 — Select and Implement the Best Alternative

Weighed against the criteria, the optimal path is a hybrid approach: assembling a small hybrid team (a lean mix of one or two experienced developers guiding a small group of trained juniors) to build and launch an MVP. This choice best balances the heavily weighted criteria of speed and cost while still allowing the company to gather real market feedback before investing further in scalability.

## 7. Conclusion & Recommendations

Taken together, the group's analysis shows that Delta Tech faces a layered decision-making challenge: routine, programmed matters such as payroll, stand-ups, and licensing can be managed through standard procedures, while the truly strategic questions—budget allocation, demand uncertainty, and talent sourcing—are non-programmed and unfold under conditions of genuine uncertainty rather than measurable risk. Because reliable data is scarce and time is short, management will necessarily lean on intuition and heuristics such as availability and satisficing, but these instincts should be paired with the discipline of the rational decision-making process wherever possible. On this basis, the group recommends that Delta Tech proceed by assembling a small hybrid development team and launching a minimum viable product rather than a full-featured application. This approach limits financial exposure, respects the tight timeline, and—most importantly—allows the company to gather real evidence of retailer demand before committing further budget. Early customer feedback should then be used to guide a data-informed pivot or scale-up decision, gradually converting an uncertain environment into one where future choices can be made with greater confidence and lower risk.